



Questions [MCQ] 2

Engineering Physics (Lovely Professional University)



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Sample MCQ

Engineering Physics: PHY110



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1. Whether the vectors $(-2, 1, -1)$ and $(0, 3, 1)$ are parallel or not
 - a. Parallel
 - b. Collinearly parallel
 - c. Not parallel
 - d. Data insufficient
 2. Find $\text{div}(\text{curl } \mathbf{F})$, where $\mathbf{F} = -x^2y\hat{i} + xz\hat{j} + 2yz\hat{k}$
 - a. 1
 - b. -1
 - c. 0
 - d. -3
 3. Find $\text{curl}(\text{grad } r^n)$, where n is constant and $\mathbf{r}$ is position vector.
 - a. 1
 - b. -1
 - c. 0
 - d. -3
 4. A field is irrotational if
 - a. $\text{grad } A = 0$
 - b. $\text{div } \mathbf{A} = 0$
 - c. $\text{Curl } \mathbf{A} = 0$
 - d. None
- Answer Key**
1.C
2. C
3.C
4.C

5. If $F = x\hat{i} + y\hat{j} + z\hat{k}$ then its divergence is

—.

- a. $\hat{i} + \hat{j} + \hat{k}$
- b. 3
- c. $x + y + z$
- d. None

7. Dielectric are the substances which are

- a. Conductor
- b. Insulator
- c. Semiconductor
- d. None

6. The Stoke's theorem is

- a. $\iint \vec{A} \cdot d\vec{S} = \oint \vec{A} \cdot d\vec{r}$
- b. $\oint \vec{A} \cdot d\vec{r} = \iint \text{curl } \vec{A} \cdot d\vec{S}$
- c. $\iint \vec{A} \cdot d\vec{S} = \iiint \text{div } \vec{A} \cdot dV$
- d. $\iint \vec{A} \cdot d\vec{S} = \iiint \text{grad } \vec{A} \cdot dV$

Answer Key

5.B

6. B

7.B

9. Maxwell's 2nd equation $\text{div } \mathbf{B} = 0$ indicates that

- a. Magnetic monopole exist
- b. Magnetic monopole doesn't exist
- c. None

10. Equation of continuity states that

- a. $\vec{\nabla} \cdot \vec{J} + \partial \rho / \partial t = 0$
- b. $\vec{\nabla} \cdot \vec{J} - \partial \rho / \partial t = 0$
- c. $-\vec{\nabla} \cdot \vec{J} + \partial \rho / \partial t = 0$
- d. None

11. Which one of these is/are correct?

- a. $\text{div } \mathbf{B} = 0$
- b. $\text{curl } \mathbf{E} = -\frac{\partial \vec{B}}{\partial t}$
- c. $\text{curl } \mathbf{B} = \frac{\partial \vec{B}}{\partial t}$
- d. Both a & b

12. The Poisson's equation in SI system is

- a. $\nabla^2 V = -\rho / \epsilon_0$
- b. $\nabla^2 V = -4\pi\rho$
- c. $\nabla^2 V = -4\pi\sigma$
- d. None

Answer Key
9.B
10. A
11.D
12.A

13. The direction of grad F is

- a. Tangential to the level surface
- b. Normal to the level surface
- c. Inclined at 45° at level surface
- d. Arbitrary

14. A field has zero divergence and zero curl.
The field is said to be

- a. divergent and rotational
- b. solenoid and rotational
- c. solenoid and irrotational
- d. divergent and irrotational

15. The Divergence's theorem is

- a. $\iint \vec{A} \cdot \vec{dS} = \oint \vec{A} \cdot \vec{dr}$
- b. $\oint \vec{A} \cdot \vec{dr} = \iint \text{curl } \vec{A} \cdot \vec{dS}$
- c. $\oint_s \vec{A} \cdot \vec{dS} = \oint_v \text{div } \vec{A} \cdot dV$
- d. $\iint \vec{A} \cdot \vec{dS} = \iiint \text{grad } A \cdot dV$

16. Which one of these is/are correct?

a. $\text{div } \mathbf{B} = 0$

b. $\text{curl } \mathbf{E} = -\frac{\partial \vec{B}}{\partial t}$

c. $\text{div } \mathbf{E} = \frac{\rho}{\epsilon_0}$

d. All

Answer Key

13.B

14. C

15.C

16.D

17. Find the divergence of the vector $F = y\hat{i} + z\hat{j} + x\hat{k}$

- a. -1
- b. 0
- c. 3
- d. 1

18. The Ampere's modified law is based on which theorem

- a. Divergence theorem
- b. Green's theorem
- c. Stoke's theorem
- d. Maxwell's theorem

19. Maxwell's 4th law satisfies that

- a. Conduction current only
- b. displacement current only
- c. Sum of conduction and displacement current
- d. None

Answer Key

17.B

18. D

19.C



Optical Fiber

1. The optical fiber is working on which principle

- a. Refraction
- b. Total internal reflection
- c. Diffraction
- d. Interference

2. A step index fiber has a core with a refractive index of 1.45 and a cladding with a refractive index of 1.40. Its numerical aperture is ____.

- a. 0.1562
- b. 0.2441
- c. 0.3775
- d. 0.4863

3. The condition for total internal reflection to take place (θ = Angle of incidence, N_1 = RI of core and N_2 = RI of cladding) is

- a. $\sin \theta \leq \frac{N_2}{N_1}$
- b. $\sin \theta \geq \frac{N_2}{N_1}$
- c. $\sin \theta = \frac{N_2}{N_1}$
- d. $\sin \theta \geq \frac{N_1}{N_2}$

4. The core of the optical fiber is

- a. Outer part of fiber
- b. Inner part of fiber
- c. Optical fiber axis
- d. None

Answer Key

1.B

2. C

3.B

4.B

5. Multi-mode fibers are

- a. Free from intermodal dispersion
- b. Suffer intermodal dispersion
- c. Partially suffer from intermodal dispersion
- d. None

6. For multimodal step index fiber

- a. RI of core is constant.
- b. RI of cladding is constant.
- c. RI of core is varied.
- d. RI of core and cladding remains constant.

7. Find the V-number of step-index fiber having a 25 μ m core radius, $n_1=1.48$, $n_2=1.46$ and wavelength = 0.82 μ m.

- a. 64.203
- b. 46.45
- c. 41.50
- d. 0

8. The numerical aperture of the fiber (n_1 = RI of core and n_2 = RI of cladding) is

- a. $\sqrt{(n_1^2 - n_2^2)}$
- b. $\sqrt{(n_1 - n_2)}$
- c. $\sqrt{(n_2^2 - n_1^2)}$
- d. $\sqrt{(n_2 - n_1)}$

Answer Key

5.B

6.D

7.B

8.A

9. The V-number of the single mode fiber is ____.

- a. $V < 2.405$
- b. $V > 2.405$
- c. $V = 2.405$
- d. None

10. The maximum number of modes supported by a graded index fiber is determined by

- a. $N_{max} > \frac{V^2}{2}$
- b. $N_{max} \cong \frac{V^2}{2}$
- c. $N_{max} < \frac{V^2}{4}$
- d. $N_{max} \cong \frac{V^2}{4}$

11. If V-number of the single mode step index fiber is 2.305, find the maximum number of supported guided mode ?

- a. 4.5042
- b. 2.6565
- c. 1.6556
- d. 1.2383

12. The sensing medium of intrinsic optical fiber sensor is

- a. Light detector
- b. Laser light
- c. Fiber
- d. None

Answer Key

9.A

10.D

11.--

12.C

13. A glass cladding fiber is made with core glass of refractive index 1.50 and the cladding is doped to give a fractional index difference 0.0005. Find the cladding index ?

- a. 1.203
- b. 1.011
- c. 1.500
- d. 1.4992

14. For multimodal graded index fiber

- a. RI of core is constant.
- b. RI of cladding is not constant.
- c. RI of core is varied.

d. RI of core and cladding remains constant

15. If V-number of the multi-mode step index fiber is 9.493, find the maximum number of supported guided mode ?

- a. 45.95
- b. 45.0
- c. 44.06
- d. 45.06

16. The refractive index of core (N_1) and cladding (N_2) of an optical fiber satisfy the relation.

- a. $N_2^2 > N_1^2$
- b. $N_2^2 < N_1^2$
- c. $N_2^2 = N_1^2$
- d. $N_2^2 \geq N_1^2$

Answer Key

13.D

14. C

15.--

16.B

Laser



1. What is the full form of LASER
 - a. Light Amplification by Spontaneous Emission of Radiation
 - b. Light Amplification by Stimulated Emission of Reaction
 - c. Light Amplification by Spontaneous Emission of Reaction
 - d. Light Amplification by Stimulated Emission of Radiation

2. The population inversion necessary for laser action used in ruby laser is

- a. electric discharge
- b. Optical pumping
- c. Direct conversion
- d. Inelastic atom-atom collision

3. A He-Ne laser is a

- a. 2-level
- b. 3-level
- c. 4-level
- d. None

Answer Key

1.D

2. B

3.C

4. Which of the following is not true for laser ?

- a. Extremely intense light
- b. Perfect monochromatic
- c. Coherent
- d. Divergent

5. If a laser operate at wavelength of 496 nm.
What is the energy of each photon in eV?

- a. 0.5 eV
- b. 2.5 eV
- c. 1 eV
- d. 1.5 eV

6. The ratio of Einstein Co-efficient A and B can be written as

- a. $(8\pi hc^3)/\nu^3$
- b. $(8\pi hc)/\nu$
- c. $(8\pi hc)/\nu^3$
- d. $(8\pi h \nu^3)/c^3$

7. Temporal coherence is

- a. Longitudinal
- b. Transverse
- c. both a & b
- d. None

Answer Key

4.D

5.--

6.D

7.A

8. Spontaneous emission of two atoms produces radiations

- a. have random phase and random direction
- b. have same phase and same direction
- c. have random phase and same direction
- d. have same phase and random direction

9. Nd: YAG Laser is

- a. 2-level
- b. 3-level
- c. 4-level
- d. None

10. Hologram is an _____ phenomenon.

- a. Dispersion
- b. Diffraction
- c. Interference
- d. None

11. Each part of hologram contains the information about

- a. Particular part of the object
- b. Entire object
- c. Important part of object
- d. Front side of object

Answer Key
8.A
9. C
10.C
11.B

12. In holographic data storage, the information is stored in _____

- a) Pendrives
- b) Cells
- c) Crystals
- d) Diode

13. The technique by which image is obtained from a hologram is called as _____

- a) Formation
- b) Construction
- c) Reconstruction
- d) Projection

14. Which of the following is used for the formation of holograms?

- a) X-ray
- b) Visible Light
- c) Infrared
- d) Lasers

15. The information in the hologram exists in _____

- a) Colored Image form
- b) Black and white image form
- c) 3-D image form
- d) Coded form

Answer Ke

12.C

13. C

14.D

15.D

16. In He-Ne Laser, the most favorable ratio of helium to Neon for satisfactory laser action is

- a. 1:4
- b. 4:1
- c. 1:7
- d. 10:1

17. Each part of hologram contains the information about

- a. Particular part of the object
- b. Entire object
- c. Important part of object
- d. Front side of object

18. GaAs Laser is

- a. Ruby laser
- b. He-Ne laser
- c. Semiconductor laser
- d. None

Answer Key
16.D
17.B
18.C